# Electric Vehicle Price Prediction in the Vietnamese Market: A Web-Scraped Dataset with LLM-Assisted Feature Extraction

Anh D. D. Hoang, Khang H. Tran, Anh N. Vu, and Nam N. Trinh

***Abstract*—The rapid growth of the electric vehicle (EV) market in Vietnam has created demand for automated price estimation tools, yet no publicly available structured dataset exists for Vietnamese EV listings. This paper presents an end-to-end pipeline for EV price prediction, from web scraping of four major Vietnamese car marketplaces to regression modeling. We introduce a two-stage LLM-assisted feature extraction approach using local (Qwen 2.5 3B) and cloud-based (GPT-5-nano) models to parse unstructured Vietnamese car listings into structured features. Through systematic data quality analysis, we identify and correct three critical issues—a 10× price inflation affecting 41% of records, systematic brand-level pricing errors, and 54.4% duplicate contamination causing test–train leakage—reducing RMSE from 1.02 billion VND (artificially deflated by leakage) to 332 million VND (honest evaluation). We benchmark four regression models (Ridge, SVR, Random Forest, XGBoost) across price segments and find that Random Forest achieves the best overall performance ($R^2 = 0.725$, RMSE $= 332$M VND), with practical accuracy for budget and mid-range segments comprising 56% of the market. Our analysis highlights that data quality remediation contributes more to prediction accuracy than model selection in this domain.**

***Index Terms*—Electric vehicle, price prediction, web scraping, large language model, feature extraction, data quality, regression, Vietnamese market.**

## I. Introduction

VIETNAM'S electric vehicle market has experienced unprecedented growth since VinFast, the country's first domestic automaker, began mass-producing EVs in 2022. With the Vietnamese government's roadmap to phase out internal combustion engine vehicles by 2040 [1] and an influx of Chinese EV imports (BYD, Wuling, Geely), the market is expanding rapidly. However, unlike mature markets in the US and Europe where structured pricing databases and valuation tools (e.g., Kelley Blue Book, AutoTrader) exist, Vietnam's EV market lacks any publicly available structured dataset or automated pricing tool.

Existing literature on car price prediction predominantly relies on clean, structured datasets sourced from dealer APIs or curated platforms such as Kaggle [2], [3]. These studies typically focus on established Western markets where listing data follows standardized schemas. The Vietnamese automotive marketplace presents fundamentally different challenges: listings are unstructured, written in Vietnamese with inconsistent formatting, distributed across multiple platforms with incompatible schemas, and subject to systematic data quality issues that are absent from curated datasets.

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Recent advances in large language models (LLMs) offer a promising approach to parsing unstructured listings into structured features. However, LLM extraction introduces its own quality concerns, including hallucinated values and systematic unit errors that propagate through the pipeline.

This paper makes four contributions:

1) We construct the first Vietnamese EV dataset by scraping and harmonizing listings from four major car marketplaces, yielding 4,166 raw records with a 20-column unified schema.
2) We design a two-stage LLM-assisted feature extraction pipeline using a local 3B-parameter model (Qwen 2.5) with cloud-based verification (GPT-5-nano), incorporating auto-retry and Pydantic-based validation.
3) We conduct systematic data quality analysis that identifies three critical issues—source-level price inflation, brand-level systematic error, and duplicate-induced test–train leakage—and demonstrate that correcting these issues reduces RMSE by 67%, from 1.02B to 332M VND.
4) We provide segment-specific evaluation showing that the model achieves practical accuracy (RMSE $<$ 100M VND) for budget and mid-range segments, which constitute 56% of the Vietnamese EV market.

## II. Related Work

### A. Car Price Prediction

Hedonic pricing models have long been used to estimate vehicle values based on observable characteristics [6]. More recently, machine learning approaches have shown improved accuracy. Gegic et al. [2] compared multiple regression models on a Bosnian used-car dataset and found that Random Forest outperformed linear models. Pal et al. [4] applied Lasso and Random Forest to Indian car prices, achieving $R^2 > 0.90$ on a clean dealer dataset. Pudaruth [5] demonstrated that ensemble methods consistently outperform single models for used car valuation in Mauritius.

A common limitation of these studies is their reliance on pre-cleaned, single-source datasets. Our work differs by constructing a multi-source dataset from heterogeneous web platforms and explicitly addressing the data quality challenges that arise from this approach.

### B. Web Scraping for Dataset Construction

Web scraping has been widely adopted for constructing research datasets in domains where structured data is unavailable [7]. In the automotive domain, Lessmann et al. [8] scraped European car listings to study depreciation curves.

TABLE I
DATA SOURCE CHARACTERISTICS

| Website | Method | Coverage | Records |
|---|---|---|---|
| bonbanh.com | BeautifulSoup | Multi-brand | 1,502 |
| chotot.com | Scrapy | Multi-brand | 1,727 |
| xevinfastluot.com | Selenium | VinFast only | 208 |
| otodien.vn | BeautifulSoup | Multi-brand | 729 |
| **Total (merged)** | | | **4,166** |

However, the challenges of scraping Vietnamese-language marketplaces—including encoding issues, inconsistent pricing units, and platform-specific data formats—remain underexplored.

### C. LLM-Based Information Extraction

The application of LLMs for structured information extraction from unstructured text has gained traction [9]. Recent work has demonstrated that even small language models (3B–7B parameters) can effectively extract structured data when combined with appropriate prompting and validation strategies [10]. Our pipeline extends this approach with a two-stage architecture and domain-specific guardrails.

## III. DATA COLLECTION

### A. Source Websites

We scraped EV listings from four Vietnamese car marketplace websites, each offering different coverage, data quality, and listing formats. Table I summarizes the source characteristics.

Each website required a different scraping strategy due to varying levels of JavaScript rendering and anti-bot measures. bonbanh.com and otodien.vn serve static HTML and were scraped using BeautifulSoup [11]. chotot.com uses dynamic loading and was handled with Scrapy [12] and custom middleware. xevinfastluot.com, a VinFast-specialized dealer portal, required Selenium [13] for browser-based interaction.

### B. Schema Harmonization

The four sources use incompatible listing schemas. We designed a unified 20-column schema covering: identification (id, brand, base_model, model_mode), specifications (year, mileage_km, body_type, seats, doors, drivetrain, origin), condition indicators (condition, has_aftermarket_mods, exterior_color), metadata (city, seller_name, post_date, website), and the target variable (price_vnd). Records from all four sources were mapped to this schema using source-specific transformation scripts.

Fig. 1 shows the distribution of records across sources after harmonization.

## IV. LLM-ASSISTED FEATURE EXTRACTION

Raw car listings from Vietnamese websites contain critical pricing features (trim level, battery capacity, drivetrain configuration) embedded in unstructured Vietnamese text. Traditional regex-based extraction fails on the diverse naming conventions across platforms.

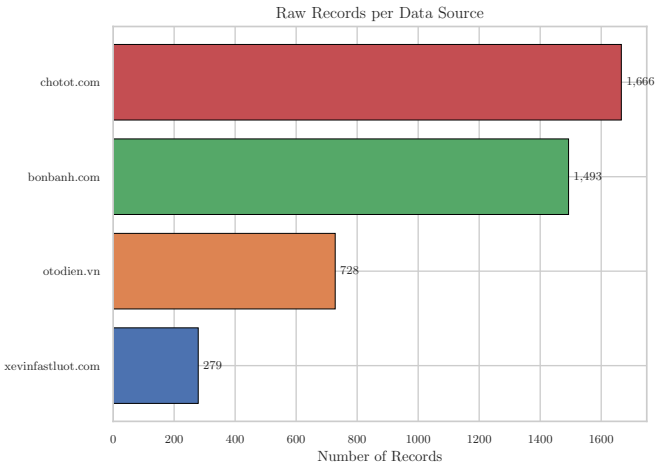


Fig. 1. Distribution of records across the four data sources.

### A. Two-Stage Pipeline

We designed a two-stage extraction pipeline:

1) **Local extraction (Qwen 2.5 3B):** Each listing's raw text is processed by a locally-hosted Qwen 2.5 3B model via Ollama [10]. The model extracts structured fields (brand, model, trim, year, condition, specifications) using a zero-shot prompt with JSON output constraints. A Pydantic schema validates the output, and failed extractions trigger automatic retry with a rephrased prompt (up to 3 attempts).
2) **Cloud verification (GPT-5-nano):** Records where the local model produced low-confidence or inconsistent results are batch-processed through OpenAI's GPT-5-nano API for verification and correction.

### B. Extraction Quality

The two-stage approach achieved high fill rates for core fields: brand (99.3%), base_model (98.7%), year (98.3%), and price (100%). The primary failure mode was trim-level extraction (model_mode), where the local model occasionally hallucinated non-existent trim names for non-VinFast brands.

## V. DATA QUALITY ANALYSIS

Systematic analysis of the merged dataset revealed three critical data quality issues, each with significant impact on model performance.

### A. chotot.com 10× Price Inflation

Cross-source price comparison revealed that listings from chotot.com exhibited prices approximately 10× higher than equivalent listings on other platforms. For example, a VinFast VF5 Plus 2024 was listed at 3.99B VND on chotot.com versus 399M VND on bonbanh.com. This affected 1,727 records (41.4% of the dataset) and was traced to a unit mismatch in the LLM extraction pipeline where prices in thousands of VND were interpreted as raw VND values.

The correction was straightforward: dividing all chotot.com prices by 10. Fig. 2 illustrates the price discrepancy before correction.

Fig. 2. Price comparison between chotot.com and other sources for the same vehicle models, showing systematic 10× inflation.

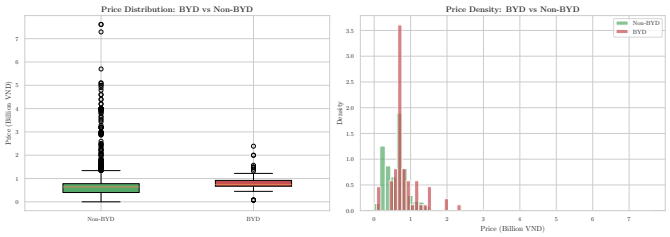


Fig. 3. Price distribution: BYD (all records 10× inflated) versus non-BYD brands.

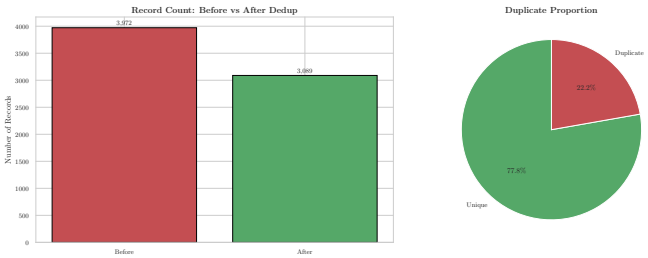


Fig. 4. Record counts before and after deduplication.

### B. BYD Systematic Pricing Error

All 74 BYD-brand records (2.3% of the dataset) exhibited prices approximately 10× higher than actual market values. Unlike the chotot.com issue, this error was brand-specific and appeared across all sources. Because every BYD record was affected, no within-brand correction was possible. Despite representing only 2.3% of records, BYD contributed 14.5% of total prediction error (MAE of 1.735B VND vs. 433M VND for non-BYD records). We removed all BYD records from the training dataset.

Fig. 3 shows the price distribution comparison between BYD and non-BYD records.

### C. Duplicate Contamination and Test–Train Leakage

Deduplication analysis on the key tuple (brand, base_model, model_mode, year, price_vnd, condition, mileage_km) revealed that 883 records (22.2% after price correction) were exact duplicates. The most extreme case—a VinFast VF9 2025 at 12.29B VND—appeared 23 times.

When duplicates span both training and test sets, the model effectively memorizes rather than generalizes, producing artificially inflated performance metrics. Before deduplication, the apparent best $R^2$ was 0.787; after proper deduplication, the honest $R^2$ was 0.725—a 7.9% reduction revealing the extent of leakage.

Fig. 4 and Fig. 5 illustrate the deduplication impact.

TABLE II
IMPACT OF DATA QUALITY CORRECTIONS

| Stage | Records | Best RMSE |
|---|---|---|
| Raw (with all issues) | 3,963 | 1.02B VND |
| + chotot.com price fix | 3,963 | – |
| + Deduplication | 3,089 | – |
| + BYD removal | 3,015 | 332M VND |

### D. Impact Summary

Table II quantifies the cumulative impact of each data quality fix.

## VI. Feature Engineering

After data cleaning, we engineered 41 features from the remaining 3,015 records through imputation, derivation, EV specification recovery, and a hybrid encoding strategy.

### A. Missing Value Imputation

Domain-informed imputation was applied to four columns:

- **Brand** (27 NaN): imputed via base_model→brand lookup (all missing-brand records were identifiable VinFast models).
- **Year** (67 NaN): median year per base_model, with global median fallback.
- **Condition**: rule-based—mileage= 0 and year≥ 2025 implies "New"; otherwise "Used."

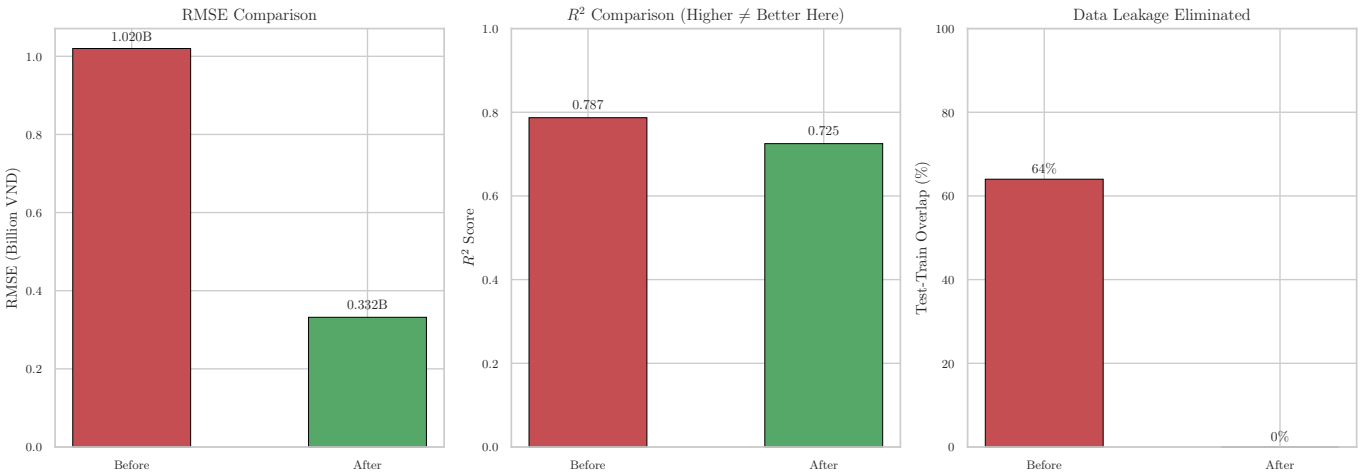


Fig. 5. Model performance before and after data quality corrections. The "before" metrics were inflated by duplicate-induced test–train leakage.

- **Mileage** (281 NaN): 0 km for new cars; median per brand for used cars. Values exceeding 300,000 km were treated as outliers and re-imputed with the per-brand median.

Vietnamese categorical values (drivetrain, origin, exterior color, city) were mapped to English equivalents for consistent encoding.

### B. Derived Features

Three numeric features were derived:

- $\texttt{car_age} = 2026 - \texttt{year}$ (depreciation proxy)
- $\texttt{log_mileage} = \log(1+\texttt{mileage_km})$ (reduces right skew)
- $\texttt{is_new} \in \{0, 1\}$ (binary condition indicator)

### C. EV Specifications Recovery

One of the most impactful feature engineering steps was recovering EV-specific technical specifications. The otodien.vn source dataset contained battery capacity (kWh), driving range (km), and motor power (hp) for 729 records with 99.6% fill rate. We constructed a lookup table mapping each base_model to its median battery_kwh, range_km, and power_hp, yielding specifications for 29 distinct EV models. This lookup was then joined to all records.

These three features capture the fundamental physics of EV pricing: a VinFast VF3 with 18.6 kWh battery and 210 km range is priced around 315M VND, while a VF9 with 92 kWh and 480 km range commands approximately 1.5B VND. Fig. 6 shows the specification distribution across models.

### D. Encoding Strategy

We adopted a hybrid encoding strategy after an important negative result: *target-encoding all categorical features caused $R^2$ to drop from 0.79 to 0.087.* The root cause was that target-encoded features for the dominant brand (VinFast, 93.6% of data) converged to a near-constant value (∼2.6B VND mean), despite actual VinFast prices ranging from 50M–14B VND. The model became "lazy," relying solely on the brand encoding.

TABLE III
FINAL FEATURE SET (41 FEATURES)

| Category | Features | Count |
|---|---|---|
| Numeric | car_age, log_mileage, is_new, has_aftermarket_mods | 4 |
| EV Specifications | battery_kwh, range_km, power_hp | 3 |
| Target-Encoded | model_mode_enc, model_year_enc | 2 |
| One-Hot: Brand | VinFast, Porsche, Mercedes, … | 5 |
| One-Hot: Model | VF3, VF5, VF6, VF7, VF8, VF9, … | 15 |
| One-Hot: Body | SUV, Hatchback, MPV, Sedan, … | 5 |
| One-Hot: Drive | AWD, FWD, RWD | 3 |
| One-Hot: Origin | Domestic, Imported, Unknown | 3 |
| **Total** | | **41** |

Our solution:

- **Target encoding** (leave-one-out with Bayesian smoothing, $m = 10$) for high-cardinality features only: `model_mode` (324 unique values, $r = 0.76$ with price) and `model_year` (interaction of base_model and year).
- **One-hot encoding** for low-cardinality features: brand ($\leq 8$), base_model (∼15), body_type (5), drivetrain (3), origin (3). Rare categories ($< 10$ records) were grouped into "Other."
- **Dropped**: exterior_color, city, brand_body (low signal after EV specs were added).

The leave-one-out target encoding formula with Bayesian smoothing is:

$$\hat{y}_i = \frac{n_c \cdot \bar{y}_c + m \cdot \bar{y}_{\text{global}}}{n_c + m} \tag{1}$$

where $n_c$ is the category count, $\bar{y}_c$ is the category mean, and $m = 10$ is the smoothing parameter. During training, each observation's own value is excluded from the category mean to prevent leakage.

Fig. 7 and Fig. 8 show the encoding strategy and resulting feature importance.

Table III summarizes the final feature set.

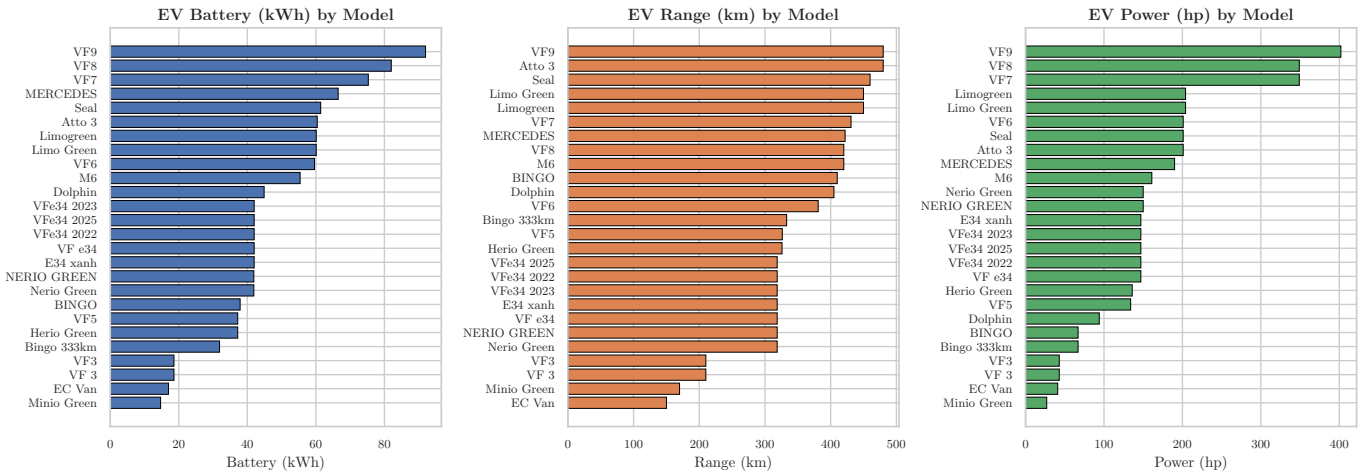


Fig. 6. EV specifications (battery capacity, range, power) across models in the lookup table.

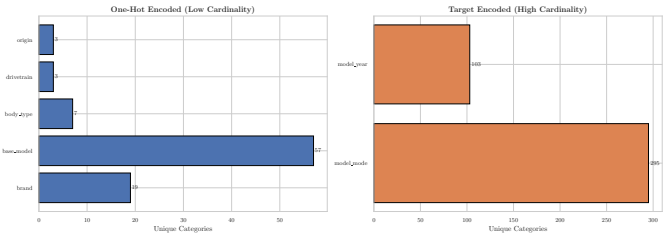


Fig. 7. Hybrid encoding strategy: one-hot for low cardinality (left), target encoding for high cardinality (right).

## VII. Experimental Setup

### A. Train–Test Split

The cleaned dataset (3,015 records) was split 80/20 into training (2,412) and test (603) sets, stratified on `base_model`. Stratification on base_model is more informative than on brand in this dataset, where 93.6% of records belong to VinFast. Rare models ($< 5$ records) were grouped for stratification purposes. Sample weights based on inverse quintile frequency of the target-encoded base_model were computed to mitigate brand concentration bias during training.

### B. Models

We benchmarked four regression models representing different paradigms:

- **Ridge Regression**: $L_2$-regularized linear model trained on standardized features with log-transformed target.
- **Support Vector Regression (SVR)**: RBF kernel on standardized features with log-transformed target.
- **Random Forest**: Ensemble (300–600 trees, grid-searched) on unscaled features with raw price target.
- **XGBoost**: Gradient-boosted trees on unscaled features with raw price target.

Ridge and SVR use log-transformed targets and standardized features to handle the heavy right-skew of Vietnamese EV prices (skewness = 2.50 for raw prices, 0.95 for log-transformed). Tree-based models operate on raw prices and unscaled features, leveraging their inherent invariance to monotonic transformations.

### C. Hyperparameter Tuning

All models were tuned via 5-fold cross-validation using GridSearchCV. The search grids were designed for the dataset characteristics (2,412 training rows, 41 features):

- Ridge: $\alpha \in \{0.001, 0.01, 0.1, 1, 10, 100\}$
- SVR: $C \in \{1, 10, 100, 1000\}$, $\epsilon \in \{0.001, 0.01, 0.1\}$, $\gamma \in \{\text{scale}, \text{auto}\}$
- RF: $n_{\text{est}} \in \{300, 600\}$, max_depth $\in \{10, 20, \text{None}\}$, max_features $\in \{\sqrt{}, 0.5, 0.8\}$
- XGBoost: $n_{\text{est}} \in \{300, 600\}$, max_depth $\in \{4, 8, 12\}$, learning_rate $\in \{0.01, 0.05, 0.1\}$, subsample $\in \{0.8, 1.0\}$, colsample_bytree $\in \{0.8, 1.0\}$, min_child_weight $\in \{3, 5\}$

### D. Evaluation Metrics

We report RMSE, MAE, $R^2$, adjusted $R^2$, and MAPE on the held-out test set. Additionally, we introduce segment-specific evaluation using four price tiers reflecting the Vietnamese EV market structure:

- **Budget**: $< 500$M VND (entry-level EVs)
- **Mid**: 500M–1.2B VND (family EVs)
- **Premium**: 1.2B–5B VND (high-end domestic)
- **Luxury**: $> 5$B VND (imported luxury)

## VIII. Results and Discussion

### A. Overall Benchmark

Table IV presents the test-set performance of all four models after hyperparameter tuning on the cleaned dataset.

Random Forest achieved the best performance across all metrics, with RMSE of 332M VND and $R^2 = 0.725$. The 95% confidence interval for RF RMSE is [139M, 496M] VND, computed via bootstrap resampling. The tree-based models (RF and XGBoost) and SVR significantly outperformed Ridge Regression in MAE, suggesting that non-linear relationships between features and price are important.

Fig. 9 and Fig. 10 compare the models visually.

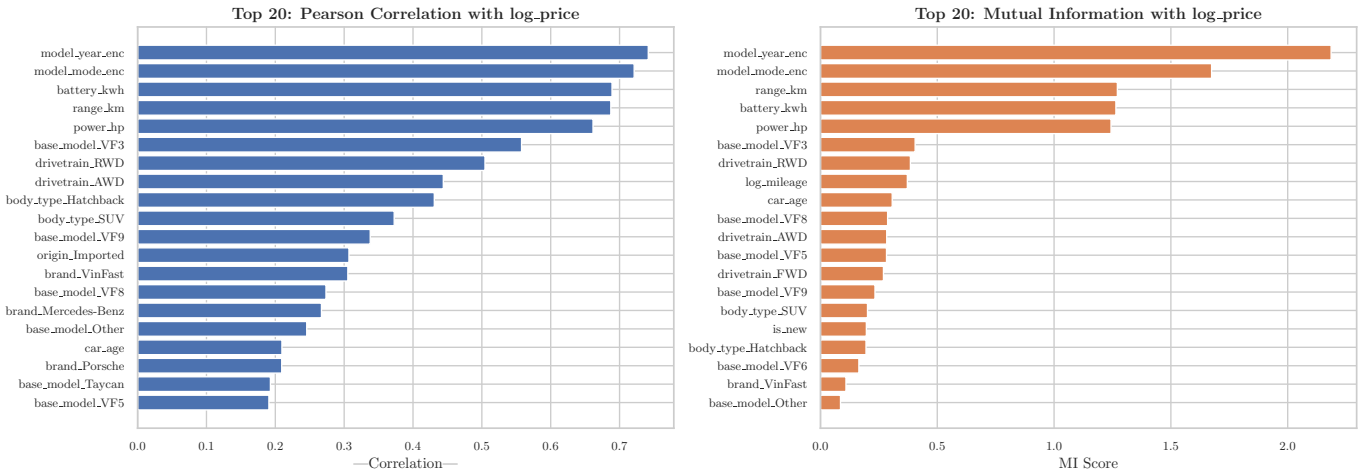


Fig. 8. Top 20 features ranked by Pearson correlation (left) and mutual information (right) with log-transformed price.

TABLE IV
MODEL BENCHMARK RESULTS ON TEST SET ($n = 603$)

| **Model** | **RMSE** (M VND) | **MAE** (M VND) | $R^2$ | **Adj.** $R^2$ | **MAPE** (%) |
|---|---|---|---|---|---|
| Ridge | 381 | 115 | 0.636 | 0.610 | 43.5 |
| SVR | 369 | 94 | 0.660 | 0.635 | 48.9 |
| **Random Forest** | **332** | **89** | **0.725** | **0.705** | **43.6** |
| XGBoost | 383 | 103 | 0.633 | 0.606 | 43.0 |

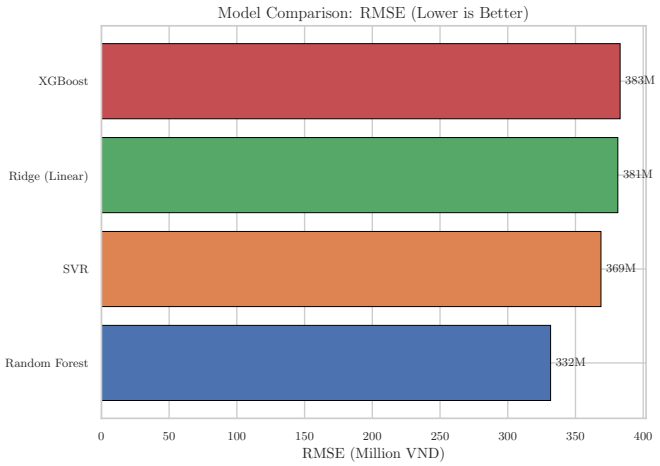


Fig. 9. RMSE comparison across four regression models.

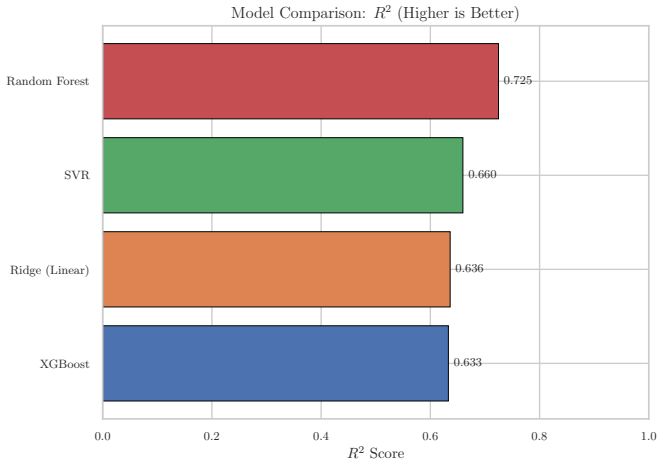


Fig. 10. $R^2$ comparison across four regression models.

TABLE V
RANDOM FOREST PERFORMANCE BY PRICE SEGMENT

| **Segment** | $n$ **(test)** | **RMSE (M VND)** | **MAE (M VND)** |
|---|---|---|---|
| Budget (< 500M) | 237 | 100 | 45 |
| Mid (500M–1.2B) | 318 | 81 | 58 |
| Premium (1.2B–5B) | 46 | 797 | 394 |
| Luxury (> 5B) | 2 | 4,041 | 3,275 |
| **Overall** | **603** | **332** | **89** |

### B. Segment-Specific Analysis

Table V reveals substantial performance variation across price segments for the best model (Random Forest).

The budget and mid-range segments—comprising 92% of test records (555 of 603)—achieve RMSE of 100M and 81M VND respectively, representing prediction errors of approximately 20% and 10% of median segment price. These error levels are practical for price estimation on the scraped websites. The premium segment (RMSE = 797M) and luxury segment (RMSE = 4B, $n = 2$) remain impractical due to sparse data and high price variance.

Fig. 11 visualizes the segment-level RMSE across all four models.

### C. Feature Importance

Fig. 12 shows the Random Forest permutation feature importance. The target-encoded `model_mode_enc` (trim level) is the most important feature, confirming our decision to recover this high-correlation ($r = 0.76$) signal via target encoding rather than discarding it due to high cardinality. The EV specification features (battery_kwh, range_km, power_hp)

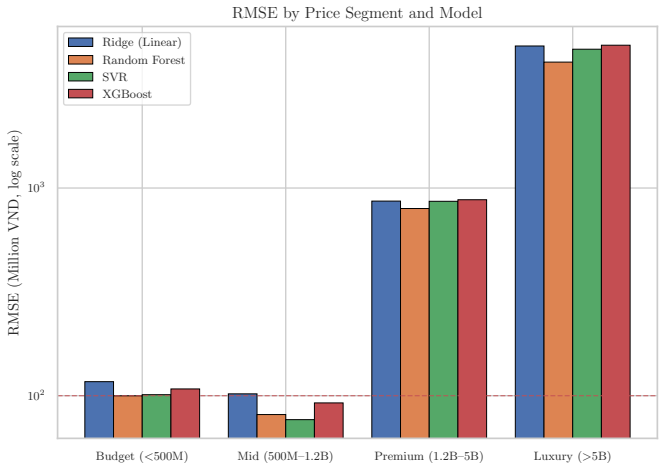


Fig. 11. RMSE by price segment for all four models. Budget and mid segments show practical accuracy; luxury is dominated by data sparsity.

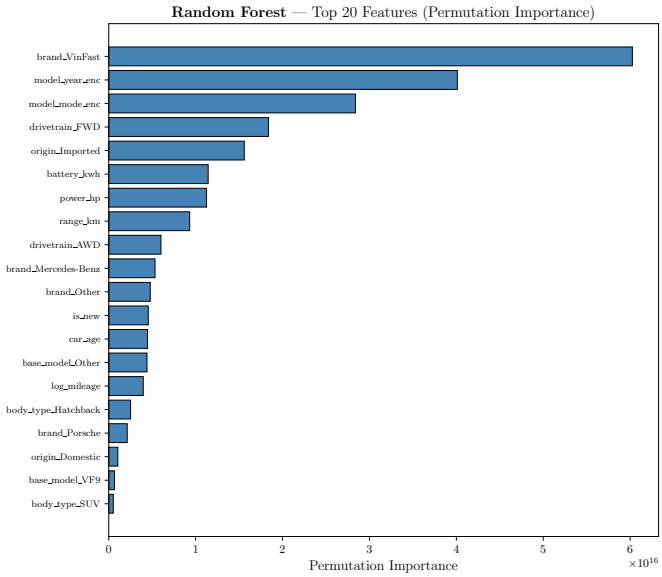


Fig. 12. Random Forest feature importance (top 20 features).

rank among the top predictors, validating the lookup-table recovery approach.

### D. Residual Analysis

Fig. 13 shows the actual versus predicted scatter for Random Forest. Predictions cluster well along the diagonal for prices below 1.5B VND, with increasing dispersion for higher-priced vehicles. Fig. 14 shows the residual distribution, which is approximately symmetric around zero with heavy tails driven by the premium and luxury segments.

### E. Data Quality vs. Model Selection

A key finding of this study is the relative importance of data quality remediation versus model architecture. The data quality corrections (chotot.com price fix, deduplication, BYD removal) reduced RMSE from 1.02B to 332M VND—a 67%

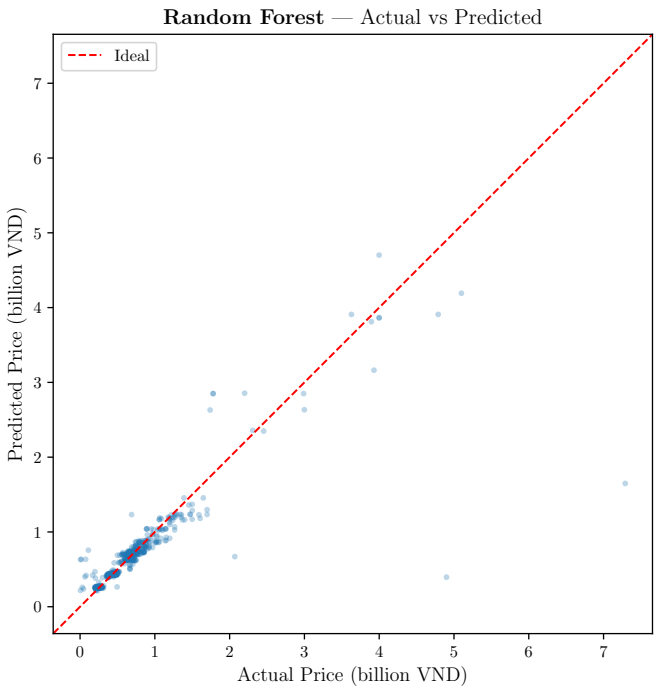


Fig. 13. Actual versus predicted prices for Random Forest. Points near the diagonal indicate accurate predictions.

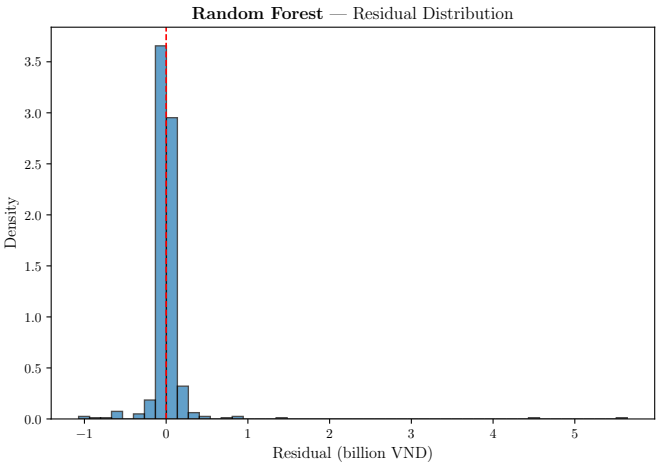


Fig. 14. Distribution of prediction residuals for Random Forest.

improvement. By contrast, the difference between the best (RF, 332M) and worst (XGBoost, 383M) models on the cleaned data is only 51M VND (15%). This suggests that in real-world applied ML settings where data is scraped rather than curated, investment in data quality analysis yields substantially higher returns than model experimentation.

## IX. Limitations and Future Work

Several limitations constrain the generalizability of our results:

**Brand concentration.** VinFast accounts for 93.6% of all records, reflecting the actual market composition but limiting the model's ability to price non-VinFast vehicles accurately.

**Dataset size.** At 3,015 clean records (2,412 training), the dataset is small by modern ML standards. This particularly affects the premium and luxury segments, where $n < 50$ in the test set.

**No temporal modeling.** All listings are treated as cross-sectional. The model does not account for time-varying factors such as depreciation curves, seasonal demand, or policy changes (e.g., EV subsidies).

**Luxury segment.** With only 2 test records in the luxury segment, the 4B VND RMSE is statistically unreliable and practically useless.

Future work could address these limitations through: (1) continuous scraping to build a time-series dataset for depreciation modeling; (2) incorporating additional data sources (official dealer prices, government registration data); (3) ensemble stacking of the four models; (4) expanding to hybrid and plug-in hybrid vehicles as the Vietnamese market diversifies.

## X. Conclusion

We presented an end-to-end pipeline for electric vehicle price prediction in the Vietnamese market, spanning web scraping, LLM-assisted feature extraction, data quality analysis, feature engineering, and regression benchmarking. Our primary finding is that systematic data quality remediation—identifying and correcting a $10\times$ source-level price inflation, removing systematically erroneous brand data, and eliminating duplicate-induced test–train leakage—reduced RMSE by 67%, from 1.02B to 332M VND. This improvement far exceeds the gains from model selection (15% difference between best and worst models on clean data).

Random Forest achieved the best overall performance ($R^2 = 0.725$, RMSE = 332M VND), with practical accuracy in the budget ($<$ 500M, RMSE = 100M) and mid-range (500M–1.2B, RMSE = 81M) segments that constitute the majority of the Vietnamese EV market. The target-encoded trim level (`model_mode_enc`) and recovered EV specifications (battery capacity, range, power) emerged as the most predictive features, confirming that domain-specific feature recovery is essential for EV pricing.

Our work demonstrates that in applied ML settings with web-scraped data, the dominant bottleneck is data quality, not model architecture—a finding with broad implications for practitioners constructing ML pipelines from real-world internet data.

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